

# Website Hosting Errors – HTTP Status Codes 300–500

## Introduction

When hosting a website, different types of errors can occur due to incorrect requests, missing files, server configuration problems, permission issues, application failures, or problems with the hosting environment.

HTTP status codes are three-digit responses returned by a web server to indicate the result of a client's request. The **300–500 range** mainly covers redirections, client-side request errors, and server-side errors.

Understanding these status codes helps administrators identify the cause of a website problem and troubleshoot it efficiently.

## HTTP Status Code Categories

| Code Range | Category      | Meaning                                                       |
|------------|---------------|---------------------------------------------------------------|
| 300–399    | Redirection   | The requested resource requires further action or redirection |
| 400–499    | Client Errors | The request from the client cannot be fulfilled               |
| 500–599    | Server Errors | The server encountered a problem while processing the request |

## 300–399: Redirection Errors

These status codes indicate that the requested resource has been moved or that the client needs to perform another action.

### 301 – Moved Permanently

#### Meaning:

The requested webpage has permanently moved to another URL.

#### Common Causes:

- Website domain has been changed.
- HTTP has been redirected to HTTPS.
- Old URL has been replaced by a new URL.
- Redirect rules configured in Apache/Nginx.

**Troubleshooting:**

- Check Apache or Nginx redirect configuration.
- Check `.htaccess` rules if Apache is being used.
- Verify the destination URL.
- Check for redirect loops.
- Confirm that the SSL/HTTPS configuration is correct.

## 302 – Found / Temporary Redirect

**Meaning:**

The requested resource is temporarily available at another URL.

**Common Causes:**

- Temporary website redirection.
- Application-generated redirects.
- Incorrect redirect configuration.
- Login or authentication redirects.

**Troubleshooting:**

- Check the web server's virtual host configuration.
- Check application redirect settings.
- Verify rewrite rules.
- Check whether multiple redirects are occurring.

## 304 – Not Modified

**Meaning:**

The requested resource has not changed since the client's cached version.

**Common Causes:**

- Browser caching.
- Server cache configuration.

- Conditional HTTP requests.

**Troubleshooting:**

- Clear browser cache.
- Check cache-control headers.
- Check web server caching configuration.
- Test the website using a private/incognito window.

**Note:** 304 is normally not a website error. It is a normal HTTP response related to caching.

## 400–499: Client-Side Errors

These errors generally occur because the client's request is invalid, unauthorized, forbidden, or pointing to a resource that does not exist.

### 400 – Bad Request

**Meaning:**

The server cannot understand or process the request.

**Common Causes:**

- Invalid request syntax.
- Incorrect URL.
- Invalid request headers.
- Corrupted cookies.
- Malformed application requests.

**Troubleshooting:**

- Verify the URL.
- Clear browser cookies and cache.
- Check request headers.
- Review application and web-server logs.
- Test the request using another browser or client.

### 401 – Unauthorized

**Meaning:**

Authentication is required or the provided credentials are invalid.

**Common Causes:**

- Incorrect username or password.
- Missing authentication credentials.
- Incorrect authentication configuration.
- Protected directory or resource.

**Troubleshooting:**

- Verify username and password.
- Check authentication configuration.
- Check Apache/Nginx configuration.
- Verify the authentication mechanism.
- Review server logs.

## **403 – Forbidden**

**Meaning:**

The server understood the request but refuses to provide access to the resource.

**Common Causes:**

- Incorrect file or directory permissions.
- Incorrect ownership.
- SELinux restrictions.
- Directory access restrictions.
- Missing index file.
- Incorrect Apache/Nginx configuration.

**Troubleshooting:**

- Check file and directory permissions.
- Check ownership.
- Check SELinux contexts and policies.
- Verify <Directory> configuration in Apache.
- Check Nginx location rules.
- Check web-server error logs.

## 404 – Not Found

### Meaning:

The requested resource does not exist at the specified URL.

### Common Causes:

- File does not exist.
- Incorrect URL.
- Incorrect DocumentRoot.
- Incorrect virtual-host configuration.
- File was moved or deleted.
- Incorrect rewrite rules.

### Troubleshooting:

- Verify that the requested file exists.
- Check the URL for spelling mistakes.
- Check the DocumentRoot.
- Check virtual-host configuration.
- Check rewrite rules.
- Review the web-server access and error logs.

## 405 – Method Not Allowed

### Meaning:

The requested HTTP method is not allowed for the resource.

### Common Causes:

- Sending POST to a resource that only accepts GET.
- Server configuration restrictions.
- Application/API configuration problems.

### Troubleshooting:

- Check the HTTP method being used.
- Check application/API configuration.
- Verify Apache/Nginx method restrictions.
- Review application logs.

## 408 – Request Timeout

### Meaning:

The server timed out while waiting for the client's request.

### Common Causes:

- Slow or unstable network.
- Client connection problems.
- Server timeout configuration.
- Overloaded server.

### Troubleshooting:

- Check network connectivity.
- Check server timeout settings.
- Review server resource usage.
- Check web-server logs.

## 409 – Conflict

### Meaning:

The request conflicts with the current state of the resource.

### Common Causes:

- Conflicting application operations.
- Concurrent modifications.
- Resource state conflicts.

### Troubleshooting:

- Check application logs.
- Verify the current state of the resource.
- Check database/application operations.

## 429 – Too Many Requests

### Meaning:

The client has sent too many requests within a specific period.

**Common Causes:**

- Rate limiting.
- Excessive API requests.
- Bots or automated requests.
- Brute-force attempts.

**Troubleshooting:**

- Check rate-limit configuration.
- Review access logs.
- Identify excessive clients or applications.
- Adjust rate limits when appropriate.
- Investigate possible automated or malicious traffic.

## 500–599: Server-Side Errors

These errors indicate that the server encountered a problem while processing a valid request.

### 500 – Internal Server Error

**Meaning:**

The server encountered an unexpected condition that prevented it from completing the request.

**Common Causes:**

- PHP/application errors.
- Incorrect server configuration.
- Incorrect `.htaccess` syntax.
- Script errors.
- Permission problems.
- Missing modules or dependencies.
- Application crashes.

**Troubleshooting:**

- Check Apache/Nginx error logs.
- Check application logs.
- Check server configuration.

- Check file permissions and ownership.
- Check PHP or application errors.
- Verify required modules are installed.
- Test the web-server configuration.

## 501 – Not Implemented

### Meaning:

The server does not support the functionality required to fulfill the request.

### Common Causes:

- Unsupported HTTP method.
- Unsupported server functionality.
- Application/server limitations.

### Troubleshooting:

- Check the requested HTTP method.
- Check web-server capabilities.
- Verify application configuration.
- Update or configure the required server functionality.

## 502 – Bad Gateway

### Meaning:

A server acting as a gateway or proxy received an invalid response from an upstream server.

### Common Causes:

- Backend application is down.
- PHP-FPM is unavailable.
- Reverse proxy configuration problems.
- Upstream server failure.
- Network communication problems between servers.

### Troubleshooting:

- Check whether the backend service is running.
- Check PHP-FPM status when applicable.

- Check Nginx/Apache proxy configuration.
- Check upstream server connectivity.
- Review web-server and application logs.

## 503 – Service Unavailable

### Meaning:

The server is temporarily unable to handle the request.

### Common Causes:

- Server overload.
- Service is stopped.
- Server maintenance.
- Application temporarily unavailable.
- Insufficient resources.

### Troubleshooting:

- Check service status.
- Check CPU and memory usage.
- Check disk space.
- Check application/backend services.
- Review server logs.
- Check whether maintenance mode is enabled.

## 504 – Gateway Timeout

### Meaning:

A gateway or proxy did not receive a response from the upstream server within the required time.

### Common Causes:

- Backend application is slow.
- Database query is taking too long.
- Network problems.
- Incorrect timeout settings.
- Overloaded backend server.

### Troubleshooting:

- Check backend application performance.
- Check database performance.
- Check network connectivity.
- Review proxy timeout settings.
- Check server resource usage.
- Review application and web-server logs.

## Important Errors to Remember

| Code | Error                 | Main Reason                    |
|------|-----------------------|--------------------------------|
| 301  | Moved Permanently     | Permanent redirect             |
| 302  | Found                 | Temporary redirect             |
| 304  | Not Modified          | Cached resource                |
| 400  | Bad Request           | Invalid client request         |
| 401  | Unauthorized          | Authentication required/failed |
| 403  | Forbidden             | Access denied                  |
| 404  | Not Found             | Resource doesn't exist         |
| 405  | Method Not Allowed    | HTTP method not permitted      |
| 408  | Request Timeout       | Request took too long          |
| 409  | Conflict              | Resource state conflict        |
| 429  | Too Many Requests     | Rate limit exceeded            |
| 500  | Internal Server Error | Server/application failure     |
| 501  | Not Implemented       | Functionality unsupported      |
| 502  | Bad Gateway           | Invalid upstream response      |
| 503  | Service Unavailable   | Server/service unavailable     |
| 504  | Gateway Timeout       | Upstream response timeout      |

## General Website Error Troubleshooting Process

When an error occurs while hosting a website, troubleshooting should generally follow this order:

### Identify the HTTP status code

Determine whether the problem is a redirection, client-side error, or server-side error.

### Check the URL

Make sure the domain, path, filename, and extension are correct.

### Check the web-server configuration

For Apache, inspect the virtual-host and directory configuration. For Nginx, inspect the server block and location configuration.

### **Check files and directories**

Verify that the requested files exist and that the web server can access them.

### **Check permissions and ownership**

Incorrect permissions can result in 403 Forbidden and sometimes contribute to application/server errors.

### **Check SELinux**

On RHEL/Rocky Linux systems, SELinux can prevent Apache or another service from accessing files even when normal Linux permissions appear correct.

### **Check required services**

Make sure services such as Apache, Nginx, PHP-FPM, MariaDB, or other application backends are running when required.

### **Check server resources**

Check CPU, RAM, disk space, and other resources when dealing with 503 or performance-related problems.

### **Check logs**

Logs are one of the most important sources of information when troubleshooting website errors.

Typical locations include:

Apache:  
/var/log/httpd/

Nginx:  
/var/log/nginx/

### **Test again after making changes**

After correcting the suspected problem, reload/restart the relevant service when necessary and test the website again.

## Conclusion

HTTP status codes provide valuable information about problems occurring during website hosting. Codes in the **300 range** generally involve redirection, **400 range** errors usually relate to the client request or access restrictions, and **500 range** errors indicate problems on the server or with backend applications.